

Experiment 1: Output

exp-1_FDS1a:

PassengerId	Survived	Pclass	...	Fare	Cabin	Embarked
0	1	0	3 ...	7.2500	NaN	S
1	2	1	1 ...	71.2833	C85	C
2	3	1	3 ...	7.9250	NaN	S
3	4	1	1 ...	53.1000	C123	S
4	5	0	3 ...	8.0500	NaN	S

[5 rows x 12 columns]

PassengerId	Survived	Pclass	...	Fare	Cabin	Embarked
0	1	0	3 ...	7.2500	NaN	S
1	2	1	1 ...	71.2833	C85	C
2	3	1	3 ...	7.9250	NaN	S
3	4	1	1 ...	53.1000	C123	S
4	5	0	3 ...	8.0500	NaN	S
...	...	...	...	...	...	...
886	887	0	2 ...	13.0000	NaN	S
887	888	1	1 ...	30.0000	B42	S
888	889	0	3 ...	23.4500	NaN	S
889	890	1	1 ...	30.0000	C148	C
890	891	0	3 ...	7.7500	NaN	Q

[891 rows x 12 columns]

Data Types

PassengerId int64

Survived int64

Pclass int64

Name str

Sex str

Age float64

SibSp int64

Parch int64

Ticket str

Fare float64

Cabin str

Embarked str

dtype: object

<class 'pandas.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

```
# Column      Non-Null Count  Dtype
---  -
```

```
0 PassengerId  891 non-null  int64
1 Survived     891 non-null  int64
2 Pclass       891 non-null  int64
3 Name         891 non-null  str
4 Sex          891 non-null  str
5 Age          714 non-null  float64
6 SibSp        891 non-null  int64
7 Parch        891 non-null  int64
8 Ticket       891 non-null  str
9 Fare         891 non-null  float64
10 Cabin       204 non-null  str
11 Embarked    889 non-null  str
```

```
dtypes: float64(2), int64(5), str(5)
```

```
memory usage: 118.9 KB
```

```
None
```

```
(891, 12)
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='str')
```

```
      PassengerId  Survived  Pclass     Age  SibSp  Parch    Fare
count  891.000000  891.000000  891.000000  714.000000  891.000000  891.000000
891.000000
mean    446.000000   0.383838   2.308642  29.699118   0.523008   0.381594  32.204208
std    257.353842   0.486592   0.836071  14.526497   1.102743   0.806057  49.693429
min      1.000000   0.000000   1.000000   0.420000   0.000000   0.000000   0.000000
25%    223.500000   0.000000   2.000000  20.125000   0.000000   0.000000   7.910400
50%    446.000000   0.000000   3.000000  28.000000   0.000000   0.000000  14.454200
75%    668.500000   1.000000   3.000000  38.000000   1.000000   0.000000  31.000000
max    891.000000   1.000000   3.000000  80.000000   8.000000   6.000000  512.329200
```

```
PassengerId    0
```

```
Survived       0
```

```
Pclass         0
```

```
Name           0
```

```
Sex            0
```

```
Age            177
```

```
SibSp          0
```

```
Parch          0
```

```
Ticket         0
```

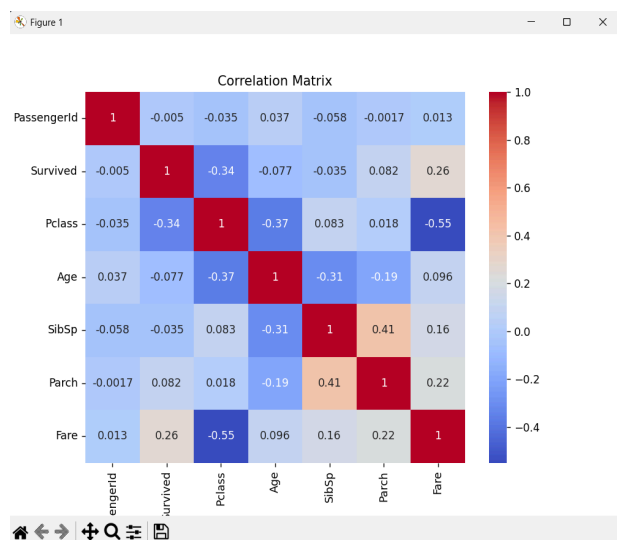
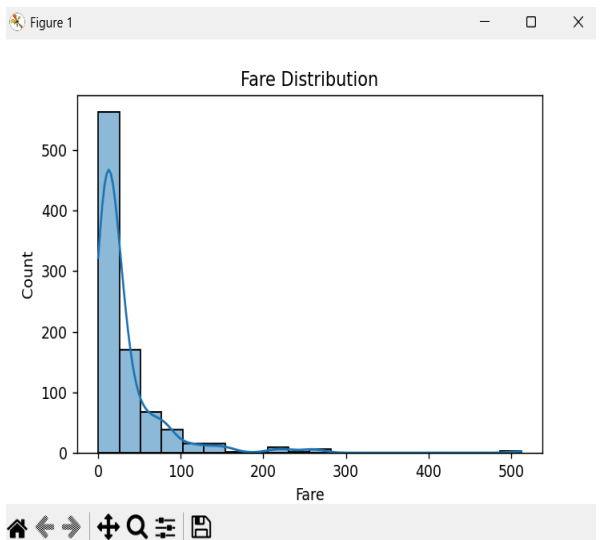
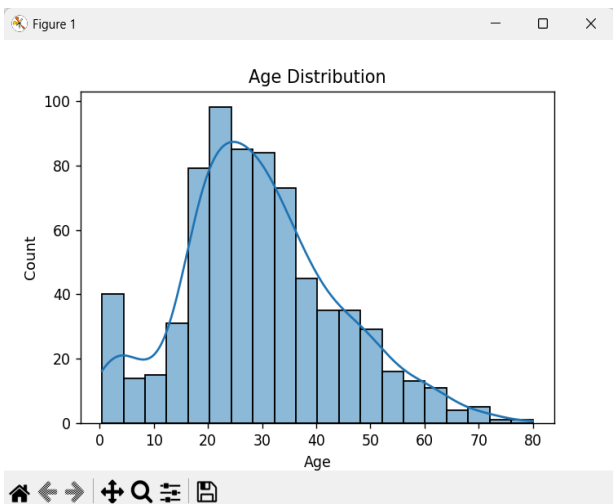
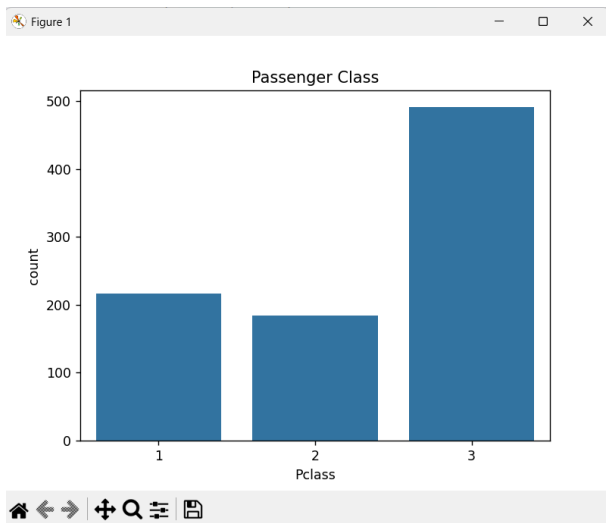
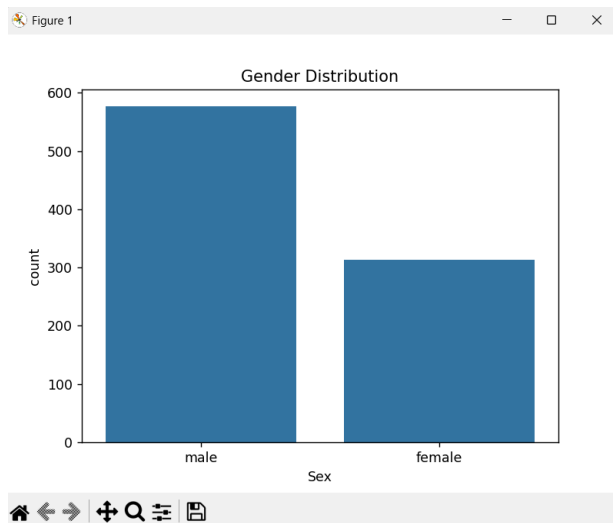
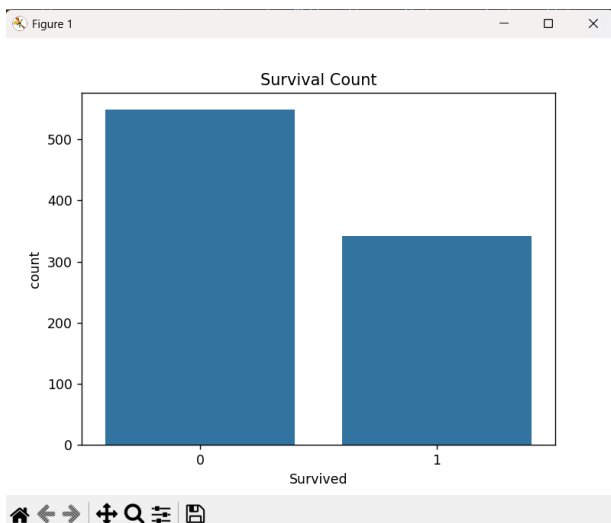
```
Fare           0
```

```
Cabin         687
```

```
Embarked       2
```

```
dtype: int64
```

exp-1_FDS1a_part2:



exp-1_FDS1b:

Numeric Attributes

Index(['PassengerId', 'Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare'], dtype='str')

Nominal Attributes

['Sex', 'Embarked']

Binary Attributes

['Survived']

exp-1_FDS1c:

Nominal Attribute Dissimilarity

Sex vs Embarked = 1.0

exp-1_FDS1d:

Euclidean Distance

66.00199624928021

Euclidean Dissimilarity

0.33998003750719785

Manhattan Distance

80.0333

manhattan Dissimilarity

0.19966700000000004

Minkowski Distance

64.36456998082116

minkowski Dissimilarity

0.3563543001917884

Cosine similarity

0.7229764648273359

Cosine Dissimilarity

0.2770235351726641